B. Tech. (Computer Engineering) Vth Semester
1st Sessional, AUTOMATA THEORY (CSC-502)

Time: 1 Hour **MM:12**

Note: Attempt all questions. **Date: 16/09/25**

Your Roll No : 23BCS066

Q No.	CO Mapped
1	1
2	1
3	2
4	2

Q. No	Question	Marks	BL
1.	A language L is defined over the alphabet {0, 1} such that L contains all strings where the number of ***0***s is divisible by 2 and does not contain the substring '***100***'. (a) Design a DFA for this language. ***Also Verify your answer using suitable examples.*** (b) How many states are needed at minimum for this Language? Justify your answer.	3+1	6
2.	Which of the following regular expressions are equivalent? ***Give reasons for your answer and also justify it using suitable examples.*** i) (00)*(Ɛ+0) ii) (00)* iii) 0(00)* iv) 0*(Ɛ+φ*)	2	4
3.	Construct a regular expression for the language L over Σ = {a, b} such that all strings contain at least one ***a***, and Do not contain the substring ***bb***. ***Verify your answer using suitable examples***.	3	3
4	Comment on the Regularity of the following language which generate strings as follows : **$L = \{ a^n b^m a^n \mid n, m >= 0 \}$**	3	4